# Amazon Beauty Reviews: NLP Analysis

7,990 valid reviews | positive=3,996 | negative=3,994

## Preprocessing rationale

Two token streams were retained for different objectives. The POS-filtered stream keeps lemmatised NOUN and ADJ tokens, reducing grammatical noise for TF-IDF and interpretable LDA topics. The lightly cleaned Word2Vec stream keeps all alphabetic content-word lemmas, preserving broader semantic context for averaged embeddings. NER runs on the original spaCy documents so entity extraction is not restricted by either modelling stream.